

GEN-3 SAFE HELICOPTER

MASTER ENGINEERING DOCUMENTS TREE

200-Document Engineering Package

Variant A (Fixed Capsule) — 100 Documents

Variant B (Ejectable Capsule with Hover Fans) — 100 Documents

10-ton / 7 × 6.0 m Rotor Baseline

Source: Document 00, Revision 00-T10-R6-Full

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PROPRIETARY CONCEPT INFORMATION

Concept / Pre-Feasibility Architecture — Not Certified

This document defines the controlled engineering-document architecture for two product variants. It does not imply that any document has been completed or verified. All numerical values remain targets or preliminary estimates pending analysis, supplier data and testing.

VARIANT A — FIXED SAFETY CAPSULE

Baseline: 10,000 kg target mass • 7 rotors × 6.0 m • 1,700 L fuel • 4 turboshaft-class prime movers (2.5–2.8 MW) • Fixed non-jettisonable capsule • 7 occupants • 8-parachute aircraft-level recovery • Ten-layer safety architecture • 17 × 17 × 4 m envelope.

00–09 | Program & Configuration Control

No.	Engineering Document	Primary Purpose
0	Master Program Baseline & Configuration Control Record	Program baseline authority
1	System Requirements Specification (SRS)	System-level requirements
2	Mission, Operational Concept & Design Reference Mission	Mission and operating definition
3	Configuration Management & Change Control Plan	Configuration governance
4	Master Data Dictionary & Parameter Control	Controlled parameter authority
5	Requirements Traceability & Verification Matrix	Bidirectional traceability
6	Engineering Evidence & Data Provenance Register	Evidence provenance
7	Interface Control Management Plan	Interface governance
8	Design Review & Engineering Gate Plan	Design-review gates
9	Engineering Maturity, Risk & Open-Issue Register	Maturity and risk tracking

10–19 | Aerodynamics, Performance & Loads

No.	Engineering Document	Primary Purpose
10	Aerodynamic Design Requirements	Aero requirements
11	Rotor Aerodynamic Model (6.0 m)	Rotor performance model
12	Hover, Climb & Forward-Flight Performance Model	Performance prediction
13	Disk Loading, Figure of Merit & Power Required Analysis	Power and efficiency
14	Stability & Control Derivatives Estimate	Stability estimates
15	External Loads & Flight Envelope Definition	Loads and envelope
16	Gust, Manoeuvre & Ground-Load Cases	Load cases
17	Noise & Vibration Preliminary Assessment	Aeroacoustic/vibration
18	Performance Validation & Flight-Test Correlation Plan	Performance verification
19	Aerodynamic Design Report (Preliminary)	Aero design synthesis

20–29 | Propulsion & Powerplant

No.	Engineering Document	Primary Purpose
20	Propulsion System Requirements	Propulsion requirements
21	Prime-Mover Selection & Matching Study (4 × turboshaft-class)	Engine selection
22	Installed Power Budget & Margin Analysis (2.5–2.8 MW target)	Power budget
23	Generator / Electrical Machine Specification	Electrical machines
24	Power Distribution Architecture (Segmented HV Bus)	Power distribution
25	Rotor Drive Motor & Inverter Specification	Rotor drives
26	Transmission / Gearbox Concept (if hybrid mechanical path)	Mechanical power path
27	Propulsion Thermal Management Requirements	Propulsion cooling
28	Propulsion System Safety & Isolation Architecture	Propulsion safety
29	Propulsion Integration & Test Plan	Propulsion verification

30–39 | Rotor System

No.	Engineering Document	Primary Purpose
30	Rotor System Requirements (7 × 6.0 m)	Rotor requirements
31	Rotor Blade Preliminary Design	Blade design
32	Hub, Pitch Control & Bearing Concept	Hub and control
33	Rotor Dynamics & Aeroelastic Assessment	Dynamics
34	Rotor Structural Loads & Fatigue Assessment	Rotor loads
35	Modular Rotor Isolation / Jettison Interface Concept	Emergency isolation
36	Rotor Health Monitoring Concept	Health monitoring
37	Rotor Ice Protection Concept (if required)	Ice protection
38	Rotor System Interface Control Document	Rotor interfaces
39	Rotor System Verification & Test Plan	Rotor verification

40–49 | Mechanical Systems & Landing Gear

No.	Engineering Document	Primary Purpose
40	Mechanical Systems Requirements	Mechanical requirements
41	Primary Landing-Leg System Design (7 mechanical damper legs)	Main landing gear
42	Landing-Leg Energy-Absorption Characterisation	Energy absorption
43	Ground Handling & Taxi Loads	Ground loads
44	Mechanical Actuation & Linkage Concept	Actuation
45	Vibration Isolation & Mount Design	Vibration isolation
46	Mechanical Fasteners, Joints & Critical Parts List	Critical parts
47	Maintainability & LRU Access Concept	Maintainability
48	Mechanical Health Monitoring Concept	Mechanical health
49	Mechanical System Verification & Test Plan	Mechanical verification

50–59 | Structures & Materials

No.	Engineering Document	Primary Purpose
50	Structural Design Requirements	Structural requirements
51	Primary Airframe Preliminary Design	Airframe structure
52	Rotor Support Structure Analysis	Rotor loads into structure
53	Capsule Shell & Multi-Angle Carbon-Fiber Roll-Cage Design	Capsule structure
54	Capsule-to-Airframe Interface & Damper Mount Design	Capsule interface
55	Landing-Leg & Energy-Absorption Structure	Landing structure
56	Material Selection & Allowables Plan	Materials
57	Finite Element Analysis (FEA) Model Plan	FEA
58	Static Strength, Ultimate Load & Buckling Analysis	Strength
59	Fatigue, Damage Tolerance & Structural Test Plan	Structural verification

60–69 | Safety Capsule, Occupant & Emergency Systems

No.	Engineering Document	Primary Purpose
60	Fixed Safety Capsule System Specification	Capsule system
61	Capsule Energy-Absorption Architecture (multi-stage)	Energy management
62	Capsule Isolation Mount / Damper Design (high-stroke)	Isolation
63	Occupant Seat & Restraint System (soft damper seats)	Occupant protection
64	Crashworthiness & Occupant Injury-Criterion Definition	Crashworthiness
65	Emergency Egress & Human-Factors Analysis	Egress
66	Fire Detection, Isolation & Suppression Architecture	Fire safety
67	Water Survival, Flotation & Ditching System	Water survival
68	Aircraft-Level Parachute Recovery System (8 canopies concept)	Emergency recovery
69	Emergency Systems Integration & Verification Plan	Emergency verification

70–79 | Flight Control, Avionics & Electrical

No.	Engineering Document	Primary Purpose
70	Flight Control System Requirements	FCS requirements
71	Distributed Rotor Control Law Architecture	Control laws
72	Fault Detection, Isolation & Reconfiguration Logic	Fault handling
73	Flight-Critical Computing & Software Architecture	Computing
74	Sensor Suite & Redundant Measurement Architecture	Sensors
75	Actuator / Motor Control Interface Specification	Actuation
76	Electrical Power, Buses & Emergency Power	Electrical
77	Avionics, Data Networks & Time Synchronization	Avionics
78	Hardware-in-the-Loop & Fault-Injection Plan	HIL verification
79	Flight-Control / Avionics Verification Plan	FCS verification

80–89 | Thermal, Fuel, Manufacturing & Maintenance

No.	Engineering Document	Primary Purpose
80	Fuel System Architecture & Tank Specification (1,700 L)	Fuel architecture
81	Fuel Containment, Venting & Fire-Safety Analysis	Fuel safety
82	Thermal Management & Cooling System	Thermal
83	Hot-Zone / Fire Propagation Analysis	Fire propagation
84	Environmental Control & Occupant Thermal Management	Occupant environment
85	Manufacturing Architecture & Design-for-Manufacture Plan	Manufacturing
86	Modular Interfaces, LRUs & Maintainability Concept	Maintainability
87	Ground Support Equipment & Tooling Concept	GSE
88	Maintenance Program & Scheduled Inspection Concept	Maintenance
89	Manufacturing & Maintenance Verification Plan	MRO verification

90–99 | Safety, Certification, Test & Program Closure

No.	Engineering Document	Primary Purpose
90	System Safety Program Plan (FHA / SSA)	Safety program
91	Functional Hazard Assessment (FHA)	Hazard assessment
92	Preliminary System Safety Assessment (PSSA)	System safety
93	Common-Cause & Common-Mode Failure Analysis	Common-mode
94	Certification Basis & Compliance Matrix (CS-27 / equivalent)	Certification
95	Test & Evaluation Master Plan (TEMP)	Test program
96	Ground Test Plan (structural, propulsion, systems)	Ground test
97	Flight Test Plan & Envelope Expansion Strategy	Flight test
98	Airworthiness Limitations & Continued Airworthiness Concept	Continued airworthiness
99	Program Closure, Lessons Learned & Rev-01 Trigger Report	Program closure

VARIANT B — EJECTABLE SAFETY CAPSULE

Baseline inherits Variant A architecture, then adds: ejectable / separable safety capsule • hidden under-capsule fans for independent hover / controlled descent • 6 capsule parachutes • 6 thick mechanical damper legs on capsule for soft landing • independent capsule energy & control provisions • separation dynamics and sequencing.

00–09 | Program & Configuration Control (Variant B)

No.	Engineering Document	Primary Purpose
0	Variant B Master Program Baseline & Configuration Control	Variant B baseline authority
1	Variant B System Requirements Specification (SRS)	Variant B system requirements
2	Variant B Mission, Operational Concept & Ejection Scenarios	Ejection mission definition
3	Variant B Configuration Management & Change Control Plan	Configuration governance
4	Variant B Master Data Dictionary & Parameter Control	Parameter authority
5	Variant B Requirements Traceability & Verification Matrix	Traceability
6	Variant B Engineering Evidence & Data Provenance Register	Evidence provenance
7	Variant B Interface Control Management Plan (incl. separation interfaces)	Interface governance
8	Variant B Design Review & Engineering Gate Plan	Design-review gates
9	Variant B Engineering Maturity, Risk & Open-Issue Register	Maturity and risk

10–19 | Aerodynamics, Performance & Separation Loads

No.	Engineering Document	Primary Purpose
10	Variant B Aerodynamic Design Requirements	Aero requirements
11	Rotor Aerodynamic Model (shared 7 × 6.0 m baseline)	Rotor performance
12	Hover, Climb & Forward-Flight Performance (pre-separation)	Performance
13	Capsule Separation Aerodynamics & Trajectory Model	Separation aero
14	Under-Capsule Fan Hover / Descent Performance Model	Capsule hover performance
15	Separation Loads, Shock & Transient Load Cases	Separation loads
16	Gust, Manoeuvre & Post-Separation Load Cases	Load cases
17	Noise & Vibration Assessment (incl. capsule fans)	Aeroacoustic
18	Performance Validation & Separation Test Correlation Plan	Performance verification
19	Variant B Aerodynamic Design Report (Preliminary)	Aero design synthesis

20–29 | Propulsion, Powerplant & Capsule Energy

No.	Engineering Document	Primary Purpose
20	Variant B Propulsion System Requirements	Propulsion requirements
21	Prime-Mover Selection & Matching (shared 4 × turboshaft baseline)	Engine selection
22	Installed Power Budget & Margin Analysis	Power budget
23	Capsule Independent Energy Source Specification	Capsule energy
24	Under-Capsule Fan Drive System Specification	Capsule fan drives
25	Capsule Power Distribution & Isolation Architecture	Capsule electrical
26	Main Aircraft Power Distribution (post-separation safe state)	Aircraft power after separation
27	Propulsion & Capsule Thermal Management Requirements	Thermal
28	Propulsion & Capsule Energy Safety / Isolation Architecture	Energy safety
29	Propulsion & Capsule Energy Integration & Test Plan	Energy verification

30–39 | Rotor System (Shared + Isolation)

No.	Engineering Document	Primary Purpose
30	Rotor System Requirements (7 × 6.0 m, shared baseline)	Rotor requirements
31	Rotor Blade Preliminary Design	Blade design
32	Hub, Pitch Control & Bearing Concept	Hub and control
33	Rotor Dynamics & Aeroelastic Assessment	Dynamics
34	Rotor Structural Loads & Fatigue Assessment	Rotor loads
35	Modular Rotor Isolation / Jettison Interface (enhanced for Variant B)	Emergency isolation
36	Rotor Health Monitoring Concept	Health monitoring
37	Rotor Ice Protection Concept (if required)	Ice protection
38	Rotor System Interface Control Document	Rotor interfaces
39	Rotor System Verification & Test Plan	Rotor verification

40–49 | Mechanical Systems, Landing Gear & Capsule Legs

No.	Engineering Document	Primary Purpose
40	Variant B Mechanical Systems Requirements	Mechanical requirements
41	Primary Aircraft Landing-Leg System (7 mechanical damper legs)	Main landing gear
42	Capsule Landing-Leg System (6 thick mechanical damper legs)	Capsule landing gear
43	Capsule Leg Energy-Absorption Characterisation	Capsule energy absorption
44	Separation Mechanism Mechanical Design	Separation mechanism
45	Vibration Isolation & Mount Design (pre- and post-separation)	Vibration isolation
46	Mechanical Fasteners, Joints & Critical Parts List	Critical parts
47	Maintainability & LRU Access Concept (incl. capsule)	Maintainability
48	Mechanical Health Monitoring Concept	Mechanical health
49	Mechanical System Verification & Test Plan (incl. separation)	Mechanical verification

50–59 | Structures & Materials (incl. Separation)

No.	Engineering Document	Primary Purpose
50	Variant B Structural Design Requirements	Structural requirements
51	Primary Airframe Preliminary Design (separation interfaces)	Airframe structure
52	Rotor Support Structure Analysis	Rotor loads
53	Ejectable Capsule Shell & Multi-Angle Carbon-Fiber Roll-Cage Design	Capsule structure
54	Capsule-to-Airframe Separation Interface Design	Separation interface
55	Landing-Leg & Energy-Absorption Structure (aircraft + capsule)	Landing structure
56	Material Selection & Allowables Plan	Materials
57	Finite Element Analysis (FEA) Model Plan (incl. separation transients)	FEA
58	Static Strength, Ultimate Load & Buckling Analysis	Strength
59	Fatigue, Damage Tolerance & Structural Test Plan	Structural verification

60–69 | Ejectable Capsule, Occupant & Emergency Systems

No.	Engineering Document	Primary Purpose
60	Ejectable Safety Capsule System Specification	Capsule system
61	Capsule Energy-Absorption Architecture (multi-stage + legs)	Energy management
62	Capsule Isolation Mount / Damper Design (pre-separation)	Isolation
63	Occupant Seat & Restraint System (soft damper seats)	Occupant protection
64	Crashworthiness & Occupant Injury-Criterion Definition	Crashworthiness
65	Emergency Egress & Human-Factors Analysis (post-landing)	Egress
66	Fire Detection, Isolation & Suppression (aircraft + capsule)	Fire safety
67	Water Survival, Flotation & Ditching System (capsule)	Water survival
68	Capsule Parachute Recovery System (6 canopies)	Capsule parachutes
69	Under-Capsule Hidden Fan System Specification (hover / descent)	Capsule hover fans

70–79 | Flight Control, Avionics, Separation Sequencing

No.	Engineering Document	Primary Purpose
70	Variant B Flight Control System Requirements	FCS requirements
71	Distributed Rotor Control Law Architecture (pre-separation)	Control laws
72	Fault Detection, Isolation & Reconfiguration Logic	Fault handling
73	Capsule Separation Sequencing & Logic Specification	Separation sequence
74	Capsule Independent Flight-Control / Descent Control	Capsule control
75	Sensor Suite & Redundant Measurement (aircraft + capsule)	Sensors
76	Electrical Power, Buses & Emergency Power (dual architecture)	Electrical
77	Avionics, Data Networks & Separation Data Links	Avionics
78	Hardware-in-the-Loop & Separation Fault-Injection Plan	HIL verification
79	Flight-Control / Separation / Capsule Verification Plan	FCS verification

80–89 | Thermal, Fuel, Manufacturing & Maintenance (Variant B)

No.	Engineering Document	Primary Purpose
80	Fuel System Architecture & Tank Specification (1,700 L baseline)	Fuel architecture
81	Fuel Containment, Venting & Fire-Safety Analysis	Fuel safety
82	Thermal Management & Cooling (aircraft + capsule systems)	Thermal
83	Hot-Zone / Fire Propagation Analysis (incl. separation)	Fire propagation
84	Environmental Control & Occupant Thermal Management	Occupant environment
85	Manufacturing Architecture & Design-for-Manufacture (dual variant)	Manufacturing
86	Modular Interfaces, LRUs & Maintainability (incl. capsule)	Maintainability
87	Ground Support Equipment & Tooling (separation system)	GSE
88	Maintenance Program & Scheduled Inspection Concept	Maintenance
89	Manufacturing & Maintenance Verification Plan	MRO verification

90–99 | Safety, Certification, Test & Program Closure (Variant B)

No.	Engineering Document	Primary Purpose
90	Variant B System Safety Program Plan (FHA / SSA)	Safety program
91	Functional Hazard Assessment (incl. separation hazards)	Hazard assessment
92	Preliminary System Safety Assessment (PSSA)	System safety
93	Common-Cause & Common-Mode Failure Analysis (dual systems)	Common-mode
94	Certification Basis & Compliance Matrix (separation credit)	Certification
95	Test & Evaluation Master Plan (TEMP) – Separation Focus	Test program
96	Ground Test Plan (structural, propulsion, separation mechanism)	Ground test
97	Flight Test Plan & Envelope Expansion (incl. separation demos)	Flight test
98	Airworthiness Limitations & Continued Airworthiness Concept	Continued airworthiness
99	Variant B Program Closure, Lessons Learned & Rev-01 Trigger	Program closure

Summary Comparison

No.	Engineering Document	Primary Purpose
0	Capsule	Fixed, non-jettisonable Ejectable / separable
1	Capsule hover / descent	None (aircraft-level only) Hidden under-capsule fans
2	Capsule parachutes	None (aircraft 8-canopy concept) 6 dedicated capsule canopies
3	Capsule landing legs	6 mechanical damper legs (integrated) 6 thick mechanical damper legs (independent)
4	Separation system	Not required Full separation mechanism + sequencing
5	Independent capsule energy	Not required Required (fans + control + recovery)
6	Document count	100 engineering documents 100 engineering documents
7	Complexity / risk	Lower – preferred first demonstrator Higher – advanced product branch
8	Certification difficulty	Lower relative to Variant B Higher (separation dynamics & dual systems)

No.	Engineering Document	Primary Purpose
0	Capsule type	Fixed, non-jettisonable / Ejectable / separable
1	Under-capsule fans	None / Yes – hidden fans for hover/descent
2	Capsule parachutes	None (aircraft 8-canopy) / 6 dedicated canopies
3	Capsule landing legs	6 damper legs (integrated) / 6 thick independent damper legs
4	Separation system	Not required / Full mechanism + sequencing logic
5	Capsule energy source	Not required / Independent energy required
6	Document count	100 / 100
7	Relative complexity	Lower / Higher
8	Preferred first build	Yes – demonstrator baseline / Advanced product branch

Recommendation: Complete Variant A document set first as the primary demonstrator baseline. Develop Variant B document set in parallel only for the additional separation, capsule-fan, capsule-parachute and independent-energy documents, re-using the shared baseline documents wherever possible.

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GEN-3 Safe Helicopter | Master Documents Tree | Variant A + Variant B

END OF MASTER DOCUMENTS TREE